Sidharth P L

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Executive Summary

AI & Data Science student and self-taught systems engineer with hands-on experience spanning operating systems, cybersecurity tooling, and full-stack product development. Founder of an enterprise software venture (Frissco Digital Ventures) and hackathon finalist (Top 6, Infosys TechZooka 2024), with a bias toward building things end-to-end rather than in isolation

Education

Rajalakshmi Institute of Technology Chennai, Tamil Nadu
B.Tech Artificial Intelligence and Data Science *2024 – 2028*

Entrepreneurial Venture

Founder & Software Engineer February 2020 – Present
Frissco Digital Ventures *Chennai, Tamil Nadu*

- Founded a software startup focused on enterprise software solutions.
- Developed ERP, CRM, and network simulation products using React, Spring Boot, PostgreSQL, and Docker.
- Designed and deployed full-stack applications from concept to production.
- Led product architecture, UI/UX design, backend development, and deployment.

Projects

ShOS - x86 Operating System Built from Scratch | *C, NASM Assembly, QEMU, Makefile* 2024

- Architected a 32-bit x86 operating system from scratch in C and NASM Assembly featuring a Multiboot-compliant bootloader, VGA text driver (80x25), and PS/2 keyboard driver.
- Implemented a RAM-based hierarchical filesystem with 20+ built-in shell commands, command history, tab completion, and a nano-like text editor with save/load support.
- Developed core kernel components: protected-mode memory management, custom system calls, interrupt handling, and debug-mode system monitoring, all tested on QEMU emulator.

ScopeIntel – Attack Surface Management Platform | *Python, FastAPI, React, Typescript, PostgreSQL, D3.js* 2025

- Built a full-stack Attack Surface Management platform using FastAPI and React (Typescript) to automate external asset discovery and infrastructure mapping for security engineers and bug bounty hunters.
- Engineered an async recon engine using Python semaphores and Nmap to parallelize the subdomain enumeration, reducing total discovery time by 80%
- Integrated multi-source enumeration (crt.sh, subfinder, SecurityTrails API) and implemented CNAME Takedown detection for unclaimed cloud resources (AWS S3, GitHub Pages, Heroku)
- Delivered interactive D3-powered force-graph attack surface visualization and automated PDF audit report generation with @react-pdf/renderer for enterprise security teams.

Technical Skills

Languages: Python, Java, Typescript, JavaScript, C, SQL, NASM Assembly
Frameworks: React, FastAPI, Node.js, Express, Socket.io, Yjs, D3.js
Databases and Backend Tools: MongoDB, PostgreSQL, REST APIs, JWT, WebSockets
Tools & platforms: Git, Linux, Docker, QEMU, PDF Report Generation, Security Recon Tooling.